

# OS-Skill-1

To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

## 1) Install Linux VM/WSL and Configure GCC

- a) Ubuntu installation,
- b) Terminal,
- c) Linux filesystem,
- d) GCC installation,
- e) Compiling and executing a C program

## 2. Linux Shell and Basic Commands

- a) pwd,

```
lahari@DESKTOP-AIC33F7:~$ pwd
/home/lahari
```

- b) ls,

```
lahari@DESKTOP-AIC33F7:~$ ls
CPrograms  OSLab  OSLab  cd  practical1  practical2  practical3
lahari@DESKTOP-AIC33F7:~$ ls -a
.  .bash_history  .bashrc  .config  .motd_shown  .sudo_as_admin_successful  OSLab  cd  practical2
.. .bash_logout  .cache  .local  .profile  CPrograms  OSLab  practical1  practical3
lahari@DESKTOP-AIC33F7:~$ ls -l
total 28
drwxr-xr-x 3 lahari lahari 4096 Jul 27 06:03 CPrograms
drwxr-xr-x 5 lahari lahari 4096 Aug 1 08:35 OSLab
drwxr-xr-x 2 lahari lahari 4096 Jul 29 07:58 OSLab
drwxr-xr-x 2 lahari lahari 4096 Jul 27 05:59 cd
drwxr-xr-x 3 lahari lahari 4096 Aug 3 06:00 practical1
drwxr-xr-x 2 lahari lahari 4096 Aug 3 07:03 practical2
drwxr-xr-x 2 lahari lahari 4096 Aug 3 13:04 practical3
lahari@DESKTOP-AIC33F7:~$ ls -lh
total 28K
drwxr-xr-x 3 lahari lahari 4.0K Jul 27 06:03 CPrograms
drwxr-xr-x 5 lahari lahari 4.0K Aug 1 08:35 OSLab
drwxr-xr-x 2 lahari lahari 4.0K Jul 29 07:58 OSLab
drwxr-xr-x 2 lahari lahari 4.0K Jul 27 05:59 cd
drwxr-xr-x 3 lahari lahari 4.0K Aug 3 06:00 practical1
drwxr-xr-x 2 lahari lahari 4.0K Aug 3 07:03 practical2
drwxr-xr-x 2 lahari lahari 4.0K Aug 3 13:04 practical3
lahari@DESKTOP-AIC33F7:~$ |
```

c) cd,

```
lahari@DESKTOP-AIC33F7:~$ cd Skill
lahari@DESKTOP-AIC33F7:~/Skill$ cd ..
lahari@DESKTOP-AIC33F7:~$ cd ~
```

d) mkdir,

```
lahari@DESKTOP-AIC33F7:~$ mkdir Skill1
```

e) rmdir

```
lahari@DESKTOP-AIC33F7:~$ rmdir Skill1
```

f) cp,

```
lahari@DESKTOP-AIC33F7:~$ cd Skill1
lahari@DESKTOP-AIC33F7:~/Skill1$ echo "Hello Linux" > file1.txt
lahari@DESKTOP-AIC33F7:~/Skill1$ cd ..
lahari@DESKTOP-AIC33F7:~$ cp -r Skill1
cp: missing destination file operand after 'Skill1'
Try 'cp --help' for more information.
lahari@DESKTOP-AIC33F7:~$ ls
CPrograms  OSLab  OSLab  Skill  Skill1  cd  practical1  practical2  practical3
```

g) mv,

```
lahari@DESKTOP-AIC33F7:~$ mkdir Documents
lahari@DESKTOP-AIC33F7:~$ echo "Operating Systems Lab" > file.txt
lahari@DESKTOP-AIC33F7:~$ ls
CPrograms  Documents  OSLab  OSLab  Skill  Skill1  cd  file.txt  practical1  practical2  practical3
lahari@DESKTOP-AIC33F7:~$ mv file.txt Documents/
lahari@DESKTOP-AIC33F7:~$ ls
CPrograms  Documents  OSLab  OSLab  Skill  Skill1  cd  practical1  practical2  practical3
lahari@DESKTOP-AIC33F7:~$ ls Documents
file.txt
```

h) rm,

```
lahari@DESKTOP-AIC33F7:~/Skill$ echo "Hello" > skill1.txt
lahari@DESKTOP-AIC33F7:~/Skill$ ls
Folder1  skill1.txt
lahari@DESKTOP-AIC33F7:~/Skill$ rm skill1.txt
lahari@DESKTOP-AIC33F7:~/Skill$ ls
Folder1
```

i) cat,

```
lahari@DESKTOP-AIC33F7:~/Skill$ pwd
/home/lahari/Skill
lahari@DESKTOP-AIC33F7:~/Skill$ ls
Folder1  file.txt
lahari@DESKTOP-AIC33F7:~/Skill$ echo "Hello Linux" > sample.txt
lahari@DESKTOP-AIC33F7:~/Skill$ ls
Folder1  file.txt  sample.txt
lahari@DESKTOP-AIC33F7:~/Skill$ cat sample.txt
Hello Linux
```

j) echo,

```
lahari@DESKTOP-AIC33F7:~/Skill$ echo Hello World
Hello World
lahari@DESKTOP-AIC33F7:~/Skill$ echo "Linux Lab" > file.txt
lahari@DESKTOP-AIC33F7:~/Skill$ echo $SHELL
/bin/bash
lahari@DESKTOP-AIC33F7:~/Skill$ echo $$
726
```

k) find,

```
lahari@DESKTOP-AIC33F7:~/Skill$ find . -name sample.txt
./sample.txt
lahari@DESKTOP-AIC33F7:~/Skill$ find . -name "*.c"
```

l) history,

```

lahari@DESKTOP-AIC33F7:~/Skill$ history
 1 wsl --status
 2 sudo apt update
 3 sudo apt install gcc
 4 #include <stdio.h>
 5 int main()
 6 {clear
 7 #include <stdio.h>
 8 int main()
 9 {    printf("Hello Operating Systems\n");
10 }
11 mkdir CPrograms
12 mkdir CPrograms cd CPrograms
13 mkdir CPrograms
14 cd CPrograms
15 sudo apt update
16 sudo apt install build-essential -y
17 sudo apt update && sudo apt upgrade -y
18 mkdir CPrograms
19 cd CPrograms
20 mkdir CPrograms
21 cd CPrograms
22 nano hello.c
23 gcc hello.c -o hello
24 ./hello
25 whoami
26 pwd
27 ls
28 uname -a
29 lscpu
30 cat process.c
31 whoami
32 pwd
33 ls
34 uname -a
35 lscpu
36 free -h
37 lsblk
38 ps
39 top
40 whoami
41 pwd
42 ls
43 ls -l
44 uname -a
45 cat
46 clear
47 whoami

```

m) shell usage

```

lahari@DESKTOP-AIC33F7:~/Skill$ echo $SHELL
/bin/bash
lahari@DESKTOP-AIC33F7:~/Skill$ echo $$
726
lahari@DESKTOP-AIC33F7:~/Skill$ bash --version
GNU bash, version 5.2.21(1)-release (x86_64-pc-linux-gnu)
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License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

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There is NO WARRANTY, to the extent permitted by law.

```

3. Git Repository and Project Structure - Git installation, git init, git add, git commit, git status, project directory organization (src, include, bin, docs)

```
lahari@DESKTOP-AIC33F7:~/Skill$ git --version
git version 2.43.0
lahari@DESKTOP-AIC33F7:~/Skill$ cd Folder1
lahari@DESKTOP-AIC33F7:~/Skill/Folder1$ pwd
/home/lahari/Skill/Folder1
lahari@DESKTOP-AIC33F7:~/Skill/Folder1$ mkdir OSSP_Project
lahari@DESKTOP-AIC33F7:~/Skill/Folder1$ cd OSSP_Project
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ mkdir src include bin docs
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ ls
bin docs include src
```

```
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/lahari/Skill/Folder1/OSSP_Project/.git/
```

```
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ ls -la
total 28
drwxr-xr-x 7 lahari lahari 4096 Aug  4 00:15 .
drwxr-xr-x 3 lahari lahari 4096 Aug  4 00:09 ..
drwxr-xr-x 7 lahari lahari 4096 Aug  4 00:15 .git
drwxr-xr-x 2 lahari lahari 4096 Aug  4 00:14 bin
drwxr-xr-x 2 lahari lahari 4096 Aug  4 00:14 docs
drwxr-xr-x 2 lahari lahari 4096 Aug  4 00:14 include
drwxr-xr-x 2 lahari lahari 4096 Aug  4 00:14 src
```

```
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ touch include/main.h
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ touch src/main.c
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ touch include/main.h
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ touch docs/README.md
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ find .
.
./include
./include/main.h
./docs
./docs/README.md
./src
./src/main.c
./bin
./.git
./.git/HEAD
./.git/info
./.git/info/exclude
./.git/hooks
./.git/hooks/prepare-commit-msg.sample
./.git/hooks/sendemail-validate.sample
./.git/hooks/post-update.sample
./.git/hooks/pre-rebase.sample
./.git/hooks/commit-msg.sample
./.git/hooks/applypatch-msg.sample
./.git/hooks/pre-push.sample
./.git/hooks/pre-receive.sample
./.git/hooks/pre-applypatch.sample
./.git/hooks/update.sample
./.git/hooks/push-to-checkout.sample
./.git/hooks/pre-commit.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/fsmonitor-watchman.sample
./.git/refs
./.git/refs/tags
./.git/refs/heads
./.git/objects
./.git/objects/info
./.git/objects/pack
./.git/branches
./.git/description
./.git/config
```

```

~/git/compag
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      docs/
      include/
      src/

nothing added to commit but untracked files present (use "git add" to track)
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git add .
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
      new file:   docs/README.md
      new file:   include/main.h
      new file:   src/main.c

lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git config --global user.name "Lahari"
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git config --global user.email "lahari809@gmail.com"
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git config --list
user.name=Lahari
user.email=lahari809@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git commit -m "Initial project structure"
[master (root-commit) cf9dad2] Initial project structure
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 docs/README.md
 create mode 100644 include/main.h
 create mode 100644 src/main.c
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ git status
On branch master
nothing to commit, working tree clean
(reverse-i-search)`: |

```

4. Shell Architecture and Makefile - User → Shell → Kernel → Hardware, Bash, process execution, writing a simple Makefile, using make

```

lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ mkdir Skill
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project$ cd Skill
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ nano hello.c
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ nano Makefile
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ gcc hello.c -o hello
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ ./hello
Hello, Makefile!
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ ls
Makefile  hello  hello.c
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ make
Makefile:3: *** missing separator.  Stop.
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ rm hello
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ ls
Makefile  hello.c
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ pwd
/home/lahari/Skill/Folder1/OSSP_Project/Skill
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ ls
Makefile  hello.c
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ nano Makefile
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ rm hello
rm: cannot remove 'hello': No such file or directory
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ make
gcc hello.c -o hello
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ ls
Makefile  hello  hello.c
lahari@DESKTOP-AIC33F7:~/Skill/Folder1/OSSP_Project/Skill$ ./hello
Hello, Makefile!

```

5. Process Creation using fork() - Process abstraction, fork(), parent and child processes, getpid(), getppid(), multiple fork() examples

#### CODE:

```

#include <stdio.h>

#include <unistd.h>

int main() {

    pid_t pid = fork();

    if (pid < 0) {

        perror("fork failed");

    } else if (pid == 0) {

        printf("Child: my PID is %d\n", getpid());

    } else {

        printf("Parent: my child's PID is %d\n", pid);

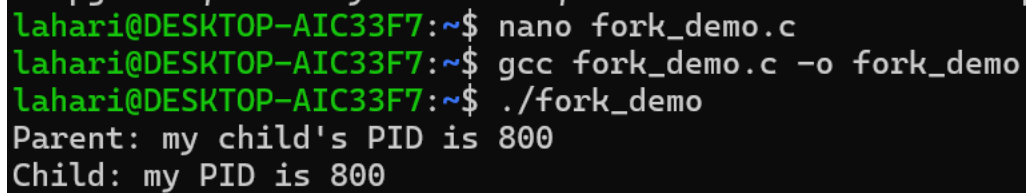
    }

}

```



```
    return 0;
}
```

A terminal window with a black background and green text. The prompt is 'lahari@DESKTOP-AIC33F7:~\$'. The user enters 'nano fork\_demo.c', then 'gcc fork\_demo.c -o fork\_demo', and finally './fork\_demo'. The output shows 'Parent: my child's PID is 800' and 'Child: my PID is 800'.

```
lahari@DESKTOP-AIC33F7:~$ nano fork_demo.c
lahari@DESKTOP-AIC33F7:~$ gcc fork_demo.c -o fork_demo
lahari@DESKTOP-AIC33F7:~$ ./fork_demo
Parent: my child's PID is 800
Child: my PID is 800
```

6. Process Replacement using the exec() Family- execl(), execv(), execvp(), replacing the current process, executing Linux commands from C

CODE:

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
int main() {
```

```
    printf("Before exec()\n");
```

```
    execlp("ls", "ls", "-l", NULL);
```

```
    // Only runs if exec fails
```

```
    perror("exec failed");
```

```
    return 1;
```

```
}
```

```

lahari@DESKTOP-AIC33F7:~$ nano exec_demo.c
lahari@DESKTOP-AIC33F7:~$ gcc exec_demo.c -o exec_demo
lahari@DESKTOP-AIC33F7:~$ ./exec_demo
Before exec()
total 84
drwxr-xr-x 3 lahari lahari 4096 Jul 27 06:03 CPrograms
drwxr-xr-x 2 lahari lahari 4096 Aug 3 23:25 Documents
drwxr-xr-x 5 lahari lahari 4096 Aug 1 08:35 OSLab
drwxr-xr-x 2 lahari lahari 4096 Aug 4 09:20 OSSP
drwxr-xr-x 2 lahari lahari 4096 Jul 29 07:58 OSlab
drwxr-xr-x 3 lahari lahari 4096 Aug 3 23:55 Skill
drwxr-xr-x 2 lahari lahari 4096 Aug 3 14:30 Skill1
drwxr-xr-x 2 lahari lahari 4096 Jul 27 05:59 cd
-rwxr-xr-x 1 lahari lahari 16048 Aug 4 09:30 exec_demo
-rw-r--r-- 1 lahari lahari 196 Aug 4 09:30 exec_demo.c
-rwxr-xr-x 1 lahari lahari 16096 Aug 4 09:25 fork_demo
-rw-r--r-- 1 lahari lahari 297 Aug 4 09:25 fork_demo.c
drwxr-xr-x 3 lahari lahari 4096 Aug 3 06:00 practical1
drwxr-xr-x 2 lahari lahari 4096 Aug 3 07:03 practical2
drwxr-xr-x 2 lahari lahari 4096 Aug 3 13:04 practical3
lahari@DESKTOP-AIC33F7:~$ |

```

7. Parent-Child Synchronization using wait(), waitpid(), synchronization, process termination order

CODE:

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main() {
```

```
    pid_t pid = fork();
```

```
    if (pid < 0) {
```

```
        perror("fork failed");
```

```
    } else if (pid == 0) {
```

```
        printf("Child (PID %d) running...\n", getpid());
```

```

    sleep(2);

    printf("Child finished\n");

    _exit(42); // exit status = 42
} else {

    int status;

    pid_t child_pid = wait(&status); // parent blocks here

    printf("Parent: child %d terminated\n", child_pid);

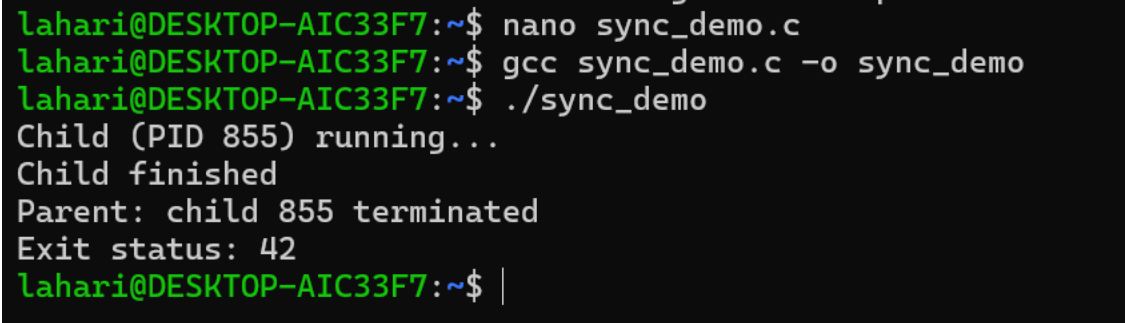
    if (WIFEXITED(status))

        printf("Exit status: %d\n", WEXITSTATUS(status));

}

return 0;
}

```



```

lahari@DESKTOP-AIC33F7:~$ nano sync_demo.c
lahari@DESKTOP-AIC33F7:~$ gcc sync_demo.c -o sync_demo
lahari@DESKTOP-AIC33F7:~$ ./sync_demo
Child (PID 855) running...
Child finished
Parent: child 855 terminated
Exit status: 42
lahari@DESKTOP-AIC33F7:~$ |

```

8. Process Tree Analysis- Multiple fork() calls, process hierarchy, pstree, ps, process IDs

CODE:

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main() {
```

```
    printf("Main process: PID = %d, Parent PID = %d\n", getpid(), getppid());
```

```

pid_t pid1 = fork(); // First fork

if (pid1 == 0) {
    // ---- Child 1 ----
    printf("Child 1: PID = %d, Parent PID = %d\n", getpid(), getppid());

    pid_t pid2 = fork(); // Second fork (inside Child 1) -> creates grandchild

    if (pid2 == 0) {
        // ---- Grandchild (Child of Child 1) ----
        printf("Grandchild: PID = %d, Parent PID = %d\n", getpid(), getppid());
    } else {
        wait(NULL); // Child 1 waits for grandchild
        printf("Child 1 (PID %d): grandchild finished\n", getpid());
    }
} else {
    // ---- Original Parent ----
    pid_t pid3 = fork(); // Third fork -> creates Child 2 (sibling of Child 1)

    if (pid3 == 0) {
        // ---- Child 2 ----
        printf("Child 2: PID = %d, Parent PID = %d\n", getpid(), getppid());
    } else {
        wait(NULL); // wait for Child 1 (or whichever finishes)
        wait(NULL); // wait for Child 2
    }
}

```

```

        printf("Parent (PID %d): all children finished\n", getpid());
    }
}

return 0;
}

```

```

lahari@DESKTOP-AIC33F7:~$ nano process_tree.c
lahari@DESKTOP-AIC33F7:~$ gcc process_tree.c -o process_tree
lahari@DESKTOP-AIC33F7:~$ ./process_tree
Main process: PID = 870, Parent PID = 458
Child 1: PID = 871, Parent PID = 870
Child 2: PID = 872, Parent PID = 870
Grandchild: PID = 873, Parent PID = 871
Child 1 (PID 871): grandchild finished
Parent (PID 870): all children finished
lahari@DESKTOP-AIC33F7:~$ |

```

9. System Call Tracing - strace, tracing ls, pwd, date, observing execve(), open(), read(), write(), close()

```

lahari@DESKTOP-AIC33F7:~$ sudo apt install strace
[sudo] password for lahari:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
strace is already the newest version (6.8-0ubuntu2).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 18 not upgraded.

```



```
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=127, ...}) = 0
mmap(NULL, 127, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efae6000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=62, ...}) = 0
mmap(NULL, 62, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efae5000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=34, ...}) = 0
mmap(NULL, 34, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efada000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=48, ...}) = 0
mmap(NULL, 48, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efad9000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=270, ...}) = 0
mmap(NULL, 270, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efad8000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=1406, ...}) = 0
mmap(NULL, 1406, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efad7000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=3360, ...}) = 0
mmap(NULL, 3360, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efad6000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=50, ...}) = 0
mmap(NULL, 50, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7042efad5000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3
```

```

shari@DESKTOP-ATC33F7:~$ strace date
execve("/usr/bin/date", ["date"], 0x7ffff93e34f00 /* 25 vars */) = 0
brk(NULL) = 0x634178698000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x73405310c000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19531, ...}) = 0
mmap(NULL, 19531, PROT_READ, MAP_PRIVATE, 3, 0) = 0x734053107000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0....", 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x734052e00000
mmap(0x734052e28000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x734052e28000
mmap(0x734052fb0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x734052fb0000
mmap(0x734052fff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x734052fff000
mmap(0x734053005000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x734053005000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x734053104000
arch_prctl(ARCH_SET_FS, 0x734053104740) = 0
set_tid_address(0x734053104a10) = 1514
set_robust_list(0x734053104a20, 24) = 0
rseq(0x734053105060, 0x20, 0, 0x53053053) = 0
mprotect(0x734052fff000, 16384, PROT_READ) = 0
mprotect(0x63415e9cc000, 8192, PROT_READ) = 0
mprotect(0x73405314c000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x734053107000, 19531) = 0
getrandom("\x51\x77\x28\x40\x80\x4e\xe5\x1b", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x634178698000
brk(0x6341786b9000) = 0x6341786b9000
openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=2996, ...}) = 0
read(3, "# Locale name alias data base.\n"... , 4096) = 2996
read(3, "", 4096) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=258, ...}) = 0
mmap(NULL, 258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x73405310b000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27028, ...}) = 0
mmap(NULL, 27028, PROT_READ, MAP_SHARED, 3, 0) = 0x7340530fd000
close(3) = 0
futext(0x73405300472c, FUTEX_WAKE_PRIVATE, 2147483647) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=23, ...}) = 0
mmap(NULL, 23, PROT_READ, MAP_PRIVATE, 3, 0) = 0x73405310a000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=47, ...}) = 0
mmap(NULL, 47, PROT_READ, MAP_PRIVATE, 3, 0) = 0x734053109000

```



[illegible]

```

+++ exited with 0 +++
lahari@DESKTOP-AIC33F7:~$ strace -e openat,read,write,close ls
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\220\243\2\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
read(3, "e 0 \ndepts /dev/pts devpts rw,"..., 1024) = 1024
read(3, "me,pagesize=2M 0 0\ndebugfs /sys/"..., 1024) = 608
read(3, "", 4096) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Locale name alias data base.\n#"..., 4096) = 2996
read(3, "", 4096) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0

```



```
#include <string.h>
```

```
#include <unistd.h>
```

```
#include <sys/types.h>
```

```
#include <sys/wait.h>
```

```
int main() {
```

```
    char command[100];
```

```
    char *args[20];
```

```
    while (1) {
```

```
        printf("mini-shell> ");
```

```
        fgets(command, sizeof(command), stdin);
```

```
        command[strcspn(command, "\n")] = '\0';
```

```
        if (strcmp(command, "exit") == 0) {
```

```
            break;
```

```
        }
```

```
        int i = 0;
```

```
        char *token = strtok(command, " ");
```

```
        while (token != NULL) {
```

```
            args[i++] = token;
```

```
            token = strtok(NULL, " ");
```

```
        }
```

```
args[i] = NULL;

pid_t pid = fork();

if (pid < 0) {
    printf("Fork failed\n");
    return 1;
}

if (pid == 0) {
    printf("Child PID: %d\n", getpid());

    execvp(args[0], args);

    printf("Command not found\n");
    exit(1);
}
else {
    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    wait(NULL);
}
}
```

```
printf("Mini shell terminated.\n");
```

```
return 0;
```

```
}
```

```
lahari@DESKTOP-AIC33F7:~$ nano mini_shell.c
lahari@DESKTOP-AIC33F7:~$ gcc mini_shell.c -o mini_shell
lahari@DESKTOP-AIC33F7:~$ ./mini_shell
mini-shell> ls
Parent PID: 5505
Child PID: 5506
Child PID: 5506
CPrograms  OSlab      cd      fork_demo  ls_trace.txt  practical2  pwd_trace.txt  sync_demo.c
Documents  Practical-10  date_trace.txt  fork_demo.c  mini_shell    practical3  shell
OSLab      Skill      exec_demo  keyboard_shell  mini_shell.c  process_tree  shell.c
OSSP       Skill1     exec_demo.c  keyboard_shell.c  practical1    process_tree.c  sync_demo
mini-shell> pwd
Parent PID: 5505
Child PID: 5507
Child PID: 5507
/home/lahari
mini-shell> date
Parent PID: 5505
Child PID: 5510
Child PID: 5510
Fri Aug 14 08:55:30 UTC 2026
mini-shell> exit
Mini shell terminated.
```